

Problem 15.29

Smith makes deposits of \$1,000 on the last day of each month in an account earning interest at rate $i^{(12)} = 0.12$. The first deposit is January 31, 2005 and the final deposit is December 31, 2029. The accumulated account is used to make monthly payments of Y starting January 31, 2030 with the final one on December 31, 2054. Find Y .

Solution

Since $i^{(12)} = 0.12$, the effective monthly interest rate is

$$j = \frac{0.12}{12} = 0.01.$$

The deposits run from January 31, 2005 through December 31, 2029, which is

$$25(12) = 300$$

monthly deposits. The withdrawals run from January 31, 2030 through December 31, 2054, which is also

$$25(12) = 300$$

monthly payments.

At December 31, 2029, the accumulated value of the deposits is

$$1000s_{\overline{300}|j}.$$

This amount funds an annuity-immediate of 300 monthly payments of Y , whose present value at December 31, 2029 is

$$Ya_{\overline{300}|j}.$$

Thus,

$$1000s_{\overline{300}|j} = Ya_{\overline{300}|j}.$$

Solving for Y ,

$$Y = 1000 \frac{s_{\overline{300}|j}}{a_{\overline{300}|j}}.$$

Since

$$s_{\overline{n}|j} = a_{\overline{n}|j}(1+j)^n,$$

we have

$$Y = 1000(1.01)^{300}.$$

Therefore,

$$\begin{aligned} Y &= 1000(1.01)^{300} \\ &\approx 19788.4663. \end{aligned}$$

Hence,

$$\boxed{Y \approx \$19,788.47}.$$